

EXERCISE—LEVEL-I

1. If $S = \{(1, 1, 0), (2, 1, 3)\} \subseteq \mathbb{R}^3$, then which one of the following vectors of $\mathbb{R}^3$ is not in the span $[S]$?
 (a) $(0, 0, 0)$ (b) $(3, 2, 3)$
 (c) $(1, 2, 3)$ (d) $(4/3, 1, 1)$
2. Which one of the following statement is correct?
 (a) There is no vector space of dimension 1.
 (b) Any three vectors of a vector space of dimension 3 are linearly independent.
 (c) There is one and only one basis of a vector space of finite dimension.
 (d) If a non-zero vector space V is generated by a finite set S , then V can be generated by a linearly independent subset of S .
3. Consider the equation $2x + 2y = 1$ and $2x - y = 1$ over $\mathbb{Z}_3$. What is the solution of (x, y) ?
 (a) $(1, 1)$ but not $(2, 0)$
 (b) $(2, 0)$ but not $(1, 1)$
 (c) Both $(1, 1)$ and $(2, 0)$
 (d) $(1/2, 0)$
4. Let $F[x]$ be the ring of polynomials in one variable x over a field F with the relation $x^n = 0$, for a fixed $n \in \mathbb{N}$.
 What is the dimension of $F[x]$ over F ?
 (a) 1 (b) $n - 1$
 (c) n (d) infinite
5. If $W_1 = \{(0, x_2, x_3, x_4, x_5) : x_2, x_3, x_4, x_5 \in \mathbb{R}\}$ and $W_2 = \{(x_1, 0, x_3, x_4, x_5) : x_1, x_3, x_4, x_5 \in \mathbb{R}\}$ be subspace of $\mathbb{R}^5$, then $\dim(W_1 \cap W_2)$ is equal to
 (a) 5 (b) 4
 (c) 3 (d) 2
6. Which one of the following is correct? The set $S = \{a + ib, c + id\}$ is a basis for the vector space $\mathbb{C}$ over $\mathbb{R}$ iff
 (a) $ad - bc = 0$ (b) $ad + bc = 0$
 (c) $ad + bc \neq 0$ (d) $ad - bc \neq 0$
7. What is the dimension of the vector space formed by the solution of the system of the following equations?
 $x_1 + x_2 + x_3 = 0, x_1 + 2x_2 = 0, x_2 - x_3 = 0$
 (a) 1 (b) 2
 (c) 3 (d) 0
8. Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of 2×2 matrices over $\mathbb{R}$ and $W_1 = \left\{ \begin{bmatrix} x & y \\ 0 & x \end{bmatrix} ; x, y \in \mathbb{R} \right\}$
 and $W_2 = \left\{ \begin{bmatrix} x & y \\ z & 0 \end{bmatrix} ; x, y, z \in \mathbb{R} \right\}$,
 then $\dim(W_1 \cap W_2)$ is equal to
 (a) 0 (b) 1
 (c) 2 (d) 3
9. What is the dimension of the vector space $\mathbb{C}$ over the field $\mathbb{R}$?
 (a) 1 (b) Infinite
 (c) 2 (d) 4

10. If V is a vector space over an infinite field F such that $\dim V = 2$, then the number of distinct subspace of V is
- (a) 2 (b) 3
(c) 4 (d) infinite
11. Which one of the following polynomials lies in the linear span of $S = \{1, 1 + x + x^2\}$?
- (a) $5x^2 + 5x + 1$ (b) $100x^2 + 10x + 10$
(c) $\pi(x^2 + 1)$ (d) $7x^2 + \pi x + 3$
12. The dimension of the subspace of $\mathbb{R}^3$ spanned by $(-3, 0, 1)$, $(1, 2, 1)$ and $(3, 0, -1)$ is
- (a) 0 (b) 1
(c) 2 (d) 3
13. Let W_1, W_2, W_3 be subspaces of a vector space V such that $W_3 \subset W_1$. Then, which one of the following is correct?
- (a) $W_1 \cap (W_2 + W_3) = W_2 + W_1 \cap W_3$
(b) $W_1 \cap (W_2 + W_3) = W_3 + W_1 \cap W_2$
(c) $W_1 \cup (W_2 + W_3) = W_2 + W_1 \cup W_3$
(d) None of these
14. Consider the real vector space $\mathbb{R}^3$. The subspace $\{(x, y, z) \in \mathbb{R}^3; y = x\}$ of $\mathbb{R}^3$ is generated by which one of the following?
- (a) $\{(1, 1, 0), (0, 0, 1)\}$ (b) $\{(1, 1, 0), (1, 0, 0)\}$
(c) $\{(1, 0, 0), (0, 1, 0)\}$ (d) $\{(1, 0, 1), (0, 0, 1)\}$
15. Let R be the set of all real number and $\mathbb{R}^2 = \{(x_1, x_2) : x_1 \in \mathbb{R}, x_2 \in \mathbb{R}\}$. Then, which one of the following is a subspace of $\mathbb{R}^2$ over $\mathbb{R}$?
- (a) $\{(x_1, x_2) : x_1 > 0, x_2 > 0\}$
(b) $\{(x_1, x_2) : x_1 \in \mathbb{R}, x_2 > 0\}$
(c) $\{(x_1, x_2) : x_1 < 0, x_2 < 0\}$
(d) $\{(x_1, 0) : x_1 \in \mathbb{R}\}$
16. Let $X = (3, 2, -1)$, $Y = (2, 4, 1)$, $Z = (4, 0, -3)$ and $W = (10, 4, -5)$ be vectors in $\mathbb{R}^3$, a real vector space. Which one of the following is correct?
- (a) $2X + Z = W, Y + Z = W$
(b) $2X - Y = Z, Y + 2Z = W$
(c) $X + Z = W, 2X + Y = Z$
(d) $Y + 2Z = W, X - Y = Z$
17. Consider the vector space V over the field of real numbers spanned by the set $S = \{(0, 1, 0, 0), (1, 1, 0, 0), (1, 0, 1, 0), (0, 0, 1, 0), (1, 1, 1, 0), (1, 0, 0, 0)\}$. What is the dimension of V ?
- (a) 1 (b) 2
(c) 3 (d) 4
18. Let V be a vector space over a field F and $\alpha \in F$ and $u \in V$. Which of the following statement is not correct?
- (a) $\alpha u = 0 \Rightarrow$ either $\alpha = 0$ or $u = 0$
(b) $|-1u| = |-1| |u|$ for all $u \in V$
(c) $\alpha 0 = 0$
(d) $0u = 0$
19. Which one of the following is correct?
- (a) R is a vector space over N .
(b) R is a vector space over $\mathbb{C}$.
(c) R is a vector space over Z .
(d) None of these
20. The Dimension of $R(Q)$ is
- (a) 1 (b) finite
(c) infinite (d) None of these
21. If W be a subspace of a finite dimensional vector space $V(F)$, then $\dim V/W$ is equal to
- (a) $\dim V - \dim W$ (b) $\dim V + \dim W$
(c) $\frac{\dim V}{\dim W}$ (d) None of these
22. Let $V_n(Z_p)$ be a vector space over the field Z_p . Then, the total number of distinct basis are
- (a) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{n!}$
(b) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{P!}$
(c) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{(n! - P!)^2}$
(d) None of these
23. Let S and T be two subspaces of R^{24} such that $\dim(S) = 19$ and $\dim(T) = 17$, then the
- (a) smallest possible value of $\dim(S \cap T)$ is 2
(b) largest possible value of $\dim(S \cap T)$ is 18
(c) smallest possible value of $\dim(S + T)$ is 19
(d) largest possible value of $\dim(S + T)$ is 22
24. Let $V_n(Z_p)$ be a vector space over the field Z_p then, the total number of subspace of dimension r are
- (a) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{(P^r - 1)(P^r - P) \dots (P^r - P^{r-1})}$
(b) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{r!}$
(c) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{P!}$
(d) $\frac{(P^n - 1)(P^n - P) \dots (P^n - P^{n-1})}{(P^n - 1)(P^n - P) \dots (P^n - P^{r-1})}$

25. The eigen vectors of matrix $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ are written in the form $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$. What is $a + b$?
- (a) 0 (b) 1/2
(c) 1 (d) 2
26. Let $S = \{(-1, 0, 1), (2, 1, 4)\}$. The value of K for which the vector $(3K + 2, 3, 10)$ belongs to the linear span of S is
- (a) -2 (b) 2
(c) 8 (d) 3
27. Let $S = \{(0, 1, \alpha), (\alpha, 1, 0), (1, \alpha, 1)\}$. Then, S is a basis for $\mathbb{R}^3$ if and only if
- (a) $\alpha \neq 0$ (b) $\alpha \neq 1$
(c) $\alpha \neq 0$ and $\alpha^2 \neq 2$ (d) $-1 \leq \alpha \leq 1$
28. If $\dim \frac{V}{W} = 8$ and $\dim W = 3$, Then, $\dim V$ is
- (a) 11 (b) 5
(c) 10 (d) 1
29. Which of the following set are subspace of P (the vector space of all polynomials)
- (a) $\{p \in P : \text{degree of } P = 4\}$
(b) $\{p \in P : \text{degree of } P \geq 3\}$
(c) $\{p \in P : \text{degree of } P \leq 4 \text{ and } P'(0) = 1\}$
(d) None of these
30. The set $\{e^{2x}, e^{3x}\}$, $x \in \mathbb{R}$, is
- (a) LI over $\mathbb{R}$
(b) LD over $\mathbb{R}$
(c) LI over any interval (a, b) , only when $0 \notin (a, b)$
(d) None of the above

Answers Level-I

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (d) | 4. (b) | 5. (d) | 6. (d) | 7. (c) | 8. (b) | 9. (c) | 10. (b) |
| 11. (b) | 12. (c) | 13. (c) | 14. (a) | 15. (d) | 16. (b) | 17. (c) | 18. (b) | 19. (b) | 20. (c) |
| 21. (a) | 22. (a) | 23. (c) | 24. (a) | 25. (b) | 26. (b) | 27. (c) | 28. (a) | 29. (d) | 30. (c) |

EXERCISE—LEVEL—II

1. Which of the following sets of vectors in $\mathbb{R}^3$ are linearly independent?
- I. $[(1, 0, 0), (0, 1, 0), (1, 1, 0)]$
 II. $[(1, 0, 0), (0, 1, 0), (0, 0, 1)]$
 III. $[(0, 1, 0), (1, 0, 1), (1, 1, 0)]$
 IV. $[(0, 0, 1), (0, 1, 0), (0, 1, 1)]$
- Select the correct answer using the codes given below
- (a) I and II (b) II and III
(c) III and IV (d) I and IV
2. Consider the real vector space $V = \mathbb{R}^3$ and following of its subsets
- I. $S = \{(x, y, z) \in V : x = y = 0\}$
 II. $T = \{(x, y, z) \in V : x = 0\}$
 III. $W = \{(x, y, z) \in V : z \neq 0\}$
- Which one of the following statement is correct?
- (a) S , T and W are subspaces
(b) Only S and W are subspaces
(c) Only T and W are subspaces
(d) Only S and T are subspaces
3. Let V be the vector space of polynomials of degree ≤ 3 ; $W_1 \subset V$ be the subspace of polynomials of degree ≤ 2 ; $W_2 \subset V$ is the subspace of polynomials vanishing at 0. Then, which of the following is not true?
- (a) Span of W_1 and W_2 is V
(b) W_1 is isomorphic to W_2
(c) $\dim(W_1 \cap W_2) = 2$
(d) $\dim W_1 < \dim W_2$
4. If V is the real vector space of all mapping from $\mathbb{R}$ to $\mathbb{R}$, $V_1 = \{f \in V | f(-x) = f(x)\}$ and $V_2 = \{f \in V | f(-x) = -f(x)\}$, then which one of the following is correct?
- (a) Neither V_1 and V_2 is a subspace of V .
(b) V_1 is a subspace of V , but V_2 is not a subspace of V .
(c) V_1 is not a subspace of V , but V_2 is a subspace of V .
(d) Both V_1 and V_2 are subspace of V .
5. The set S_1
- $$= \left\{ \alpha = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & -1 \end{bmatrix}, \beta = \begin{bmatrix} 2 & -4 & 8 \\ 6 & 0 & -2 \end{bmatrix} \right\} \text{ and } S_2 = \{f = u^3 + 3u + 4, g = u^3 + 4u + 3\} \text{ are}$$
- (a) Both linearly dependent
(b) Both linearly independent
(c) S_1 is linearly dependent but S_2 is not
(d) S_2 is linearly dependent but S_1 is not
6. Let V be a vector space over the field F of dimension n . Consider the following statements
- I. Every subset of V containing n elements is a basis of V .
 II. No linearly independent subset of V contains more than n elements.
- Which of the above statement is/are correct?
- (a) I only (b) II only
(c) Both I and II (d) Neither I nor II

7. In $\mathbb{R}^3$, consider the following three statement about the subset
 $E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1), (1, 1, 0)\}$
 I. E is linearly dependent set.
 II. Any three vectors of E are linearly independent.
 III. Any four vectors of E are linearly dependent.
 Which of the above statement are correct?
 (a) I, II and III (b) I and II
 (c) I and III (d) II and III
8. If $W_1 = \{(0, a_2, a_3)\}$ and $W_2 = \{b_1, 0, b_3\}$ are two subspace of F^3 , where F^n is a vector space of dimension n . Then, the subspace $W_1 \cap W_2$ is isomorphic to the vector space
 (a) F^1 (b) F^2
 (c) F^3 (d) $2F^4$
9. Let $S = \{(1, i, 0), (2i, 1, 1), (0, 1 + i, 1 - i)\}$ be a subset of the complex vector space $\mathbb{C}^3$ and $T = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ be a subset of the real vector space $\mathbb{R}^3$. Which of the following statements is correct?
 (a) S and T are both basis.
 (b) S is a basis but T is not a basis.
 (c) S is not a basis but T is a basis.
 (d) Neither S nor T is a basis.
10. If the set of all triples (x_1, x_2, x_3) of real number R form a vector space V_3 , then a subspace denoted by a vertical plane $y = x$ can be obtained by a linear combination of the sets.
 (a) $(1, 1, 0)$ and $(0, 0, 1)$
 (b) $(1, 1, 0)$ and $(1, 0, 0)$
 (c) $(1, 0, 0)$ and $(0, 1, 0)$
 (d) $(1, 0, 1)$ and $(0, 0, 1)$
11. Let W_1 and W_2 be finite dimensional subspace of a vector space V . If $\dim W_1 = 2$, $\dim W_2 = 2$, $\dim (W_1 + W_2) = 3$, then $\dim (W_1 \cap W_2)$ is
 (a) 1 (b) 2
 (c) 3 (d) 4
12. In the vector space $\mathbb{R}^3$, the coordinate vector of $(3, 5, -2)$ relative to the ordered basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is
 (a) $(5, 3, -2)$ (b) $(3, 5, -2)$
 (c) $(-2, 5, 3)$ (d) $(-2, 3, 5)$
13. The zero vector in the vector space R^4 is
 (a) $(0, 0)$ (b) $(0, 0, 0)$
 (c) $(0, 0, 0, 0)$ (d) None of these
14. The dimension of the vector space spanned by $(1, -2, 3, -1)$ and $(1, 1, -2, 3)$ is
 (a) 1 (b) 2
 (c) 4 (d) None of these
15. If $(m, 3, 1)$ is a linear combination of vectors $(3, 2, 1)$ and $(2, 1, 0)$ in $\mathbb{R}^3$, then the value of m is
 (a) 1 (b) 3
 (c) 5 (d) None of these
16. The set $V = \{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$ is
 (a) a vector subspace of $\mathbb{R}^2$
 (b) not a vector subspace of $\mathbb{R}^2$, since every element does not have an inverse in V
 (c) not a vector subspace of $\mathbb{R}^2$, since it is not closed under scalar multiplication
 (d) not a vector subspace of $\mathbb{R}^2$, since it is not closed under vector addition
17. The set of all $x \in \mathbb{R}$ for which the vectors $(1, x, 0)$, $(0, x^2, 1)$ and $(0, 1, x)$ are linearly independent in $\mathbb{R}^3$ is
 (a) $\{x \in \mathbb{R} : x = 0\}$ (b) $\{x \in \mathbb{R} : x \neq 0\}$
 (c) $\{x \in \mathbb{R} : x \neq 1\}$ (d) $\{x \in \mathbb{R} : x \neq -1\}$
18. $S = \{(x_1, x_2, \dots, x_{100}) \in \mathbb{R} : \text{s.t. } x_1 = x_2 = \dots = x_{50}, x_{51} + x_{52} + \dots + x_{100} = 0\}$. Then, $\dim S$ is
 (a) 49 (b) 50
 (c) 47 (d) 51
19. If V and W are subspaces of $\mathbb{R}^n$, then
 (a) $V \cup W$ is necessarily a subspace of $\mathbb{R}^n$
 (b) $V \cup W$ is never a subspace of $\mathbb{R}^n$
 (c) $V \cup W$ is a subspace of $\mathbb{R}^n$ if and only if one of V, W is contained in the other
 (d) $V \cup W$ is a subspace of $\mathbb{R}^n$ if and only if one of V, W in $\{0\}$.
20. W is a 3-dimensional subspace of a finite dimensional vector space V . If the quotient space V/W has dimension 8, the dimension of V is
 (a) 3 (b) 8
 (c) 11 (d) 5
21. Let V be the vector space of all 5×5 real skew-symmetric matrices. Then, the dimension of V is
 (a) 20 (b) 15
 (c) 10 (d) 5
22. Consider the subspace $W = \{[a_{ij}] : a_{ij} = 0 \text{ if } i \text{ is even}\}$ of all 10×10 matrices. Then, dimension of W is
 (a) 25 (b) 50
 (c) 75 (d) 100
23. Let U and W be the following subspace of $\mathbb{R}^4$
 $U = \{(a, b, c, d) : b + c + d = 0\}$
 $W = \{(a, b, c, d) : a + b = 0, c = 2d\}$
 Then, $\dim$ of U, W and $U \cap W$ are, respectively
 (a) 2, 3, 1 (b) 3, 2, 1
 (c) 2, 2, 2 (d) 1, 2, 3
24. Let u, v, w are LI vectors. Then, consider the statements
 I. $u + v - 2w, u - v - w$ and $u + w$ are LI
 II. $u + v - 3w, u + 3v - w$ and $v + w$ are LD
 Choose the correct code
 (a) I is true but II is false
 (b) I is false but II is true
 (c) both I and II are true
 (d) both I and II are false

25. If $S = \{(1, 1, 0), (2, 1, 3)\} \subset \mathbb{R}^3$, then which one of the following vectors of $\mathbb{R}^3$ is not in the span $\langle S \rangle$?
- (a) $(0, 0, 0)$ (b) $(3, 2, 3)$
(c) $(1, 2, 3)$ (d) $(4/3, 1, 1)$
26. Which of the following sets of vectors is a basis for $\mathbb{R}^3$
- I. $\{(1, 2, 3), (3, 5, 7), (5, 8, 11)\}$
II. $\{(1, 0, 1), (0, 1, 0), (-1, 0, 1)\}$
III. $\{(1, 2, 3), (2, 3, 4), (2, 4, 6)\}$
- (a) only I and II (b) only II
(c) only I and III (d) only I
27. Let $S = \{(1, 2, 3), (1, 0, -1)\}$. The value of k for which the vector $(2, 1, k)$ belongs to the linear span of S is
- (a) 1 (b) -1
(c) 2 (d) 0
28. Let V be the space of $n \times n$ matrices over $\mathbb{R}$ and let W be the subspace of matrices with entries in each row adding up to zero. Then, the dimension of W is
- (a) n (b) $\frac{n(n-1)}{2}$
(c) $n-1$ (d) $n(n-1)$
29. Which of the following sets of vectors in $\mathbb{R}^3$ are linearly independent
- I. $[(1, 0, 0), (0, 1, 0), (1, 1, 0)]$
II. $[(1, 0, 0), (0, 1, 0), (0, 0, 1)]$
III. $[(0, 1, 0), (1, 0, 1), (1, 1, 0)]$
IV. $[(0, 0, 1), (0, 1, 0), (0, 1, 1)]$
- Select the correct answer
- (a) I and II (b) II and III
(c) III and IV (d) I and IV
30. The dimension of the vector space of all 3×3 real symmetric matrices is
- (a) 3 (b) 9
(c) 6 (d) 4
31. The dimension of the subspace $\{(x_1, x_2, x_3, x_4, x_5) : 3x_1 - x_2 + x_3 = 0\}$ of $\mathbb{R}^5$ is
- (a) 1 (b) 2
(c) 3 (d) 4
32. Let n be a positive integer and let H_n be the space of all $n \times n$ matrices $A = [a_{ij}]$ with entries in $\mathbb{R}$ satisfying $a_{ij} = a_{\delta_s}$ whenever $i + j = \delta + s$ ($i, j, \delta, s = 1, 2, \dots, n$). Then, the dimension of H_n , as a vector space over $\mathbb{R}$ is
- (a) n^2 (b) $n^2 - n + 1$
(c) $2n + 1$ (d) $2n - 1$
33. Consider the set of vectors (columns) defined by $X = \{X^T \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0, \text{ where } X^T = [x_1, x_2, x_3]^T\}$. Which of the following is true?
- (a) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X .
(b) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X .
(c) X is not a subspace for $\mathbb{R}^3$.
(d) None of the above
34. Let V be the vector space of real polynomials of degree not exceeding 2. Let $f(x) = x - 1$, $g(x) = x + 1$, $h(x) = x^2 - 1$, $f(x) = x^2 + 1$. Then, the set $\{f, g, h, j\}$ is
- (a) linearly independent
(b) linearly dependent because $fg = h$
(c) linearly dependent because $f + g - h = 0$
(d) linearly dependent because $f - g - h + j = 0$
35. Consider $S = \{a_0 + a_1x + a_2x^2 + a_3x^3 : a_i \in \mathbb{R}, i = 0, 1, 2, 3\}$ with usual addition and multiplication
- (a) S does not form a vector space
(b) form vector space with $\dim 4$
(c) S does not form vector space because it does not have identity
(d) form a vector space with $\dim 3$
36. Let V and W be subspaces of $\mathbb{R}^n$. Then,
- (a) $\dim(V + W)$ must be $\dim V + \dim W$
(b) $\dim(V + W) > \dim V + \dim W$
(c) $\dim(V + W) < \min(\dim V, \dim W)$
(d) $\dim(V + W) \geq \max\{\dim V, \dim W\}$
37. Let $\{e_1, e_2, e_3\}$ be a basis of a vector space V over $\mathbb{R}$. Consider the following sets
- $A = \{e_2, e_1 + e_2, e_1 + e_2 + e_3\}$
 $B = \{e_1, e_1 - e_2, e_1 - e_2 + e_3\}$
 $C = \{2e_1, 3e_2 + e_3, 6e_1 + 3e_2 + e_3\}$, then
- (a) A and B are basis of V
(b) A and C are basis of V
(c) B and C are basis of V
(d) Only B is a basis of V
38. Which of the following sets is not a subspace of $C(a, b)$
- (a) $\left\{ f \in C(a, b) : \lim_{x \rightarrow \frac{a+b}{2}} f(x) = 0 \right\}$
(b) $\left\{ f \in C(a, b) : \lim_{x \rightarrow \frac{a+b}{2}} f(x) = 1 \right\}$
(c) $\{f \in C(-\infty, \infty) : f(x) = f(-x)\}$
(d) All of the above
39. Which of the following sets are subspaces of the space $C(a, b)$ of all real-valued continuous functions defined on the open interval (a, b)
- (a) $\{f \in C(a, b) : f(x_0) = 0, x_0 \in (a, b)\}$
(b) $\{f \in C(a, b) : f'(x) \neq 0, \text{ for all } x \in (a, b)\}$
(c) $\left\{ f \in C(a, b) : f\left(\frac{a+b}{2}\right) = 1 \right\}$
(d) None of the above
40. Which of the following is not a vector subspace of the vector space of polynomials with real coefficient
- (a) W consists of all polynomials divisible by x
(b) $W = \{p(x) \in V : p(3) = 0\}$
(c) $W = \{p(x) \in V : p(a) = p(1-a), a \in \mathbb{R}\}$
(d) W consists of all polynomials with integral coefficient

41. Let V be the vector space of all $n \times n$ matrices over R . Then, the set W , consisting of all matrices which commute with a given matrix M i.e., $W = \{M \in V : AM = MA\}$
- (a) is a subspace of V
 - (b) is not a subspace of V as it is not closed wrt vector addition
 - (c) is not a subspace of V as it is not closed wrt scalar multiplication
 - (d) None of the above
42. Let V be the space of 2×2 matrices over R and let W be the subspace generated by $\begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}$, $\begin{bmatrix} 2 & -4 \\ -5 & 7 \end{bmatrix}$ and $\begin{bmatrix} 1 & -7 \\ -5 & 1 \end{bmatrix}$. Then, dimension of W is
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
43. Consider the set of all functions f defined on $[0, 1]$ such that
- I. $f(1/2) = 0$
 - II. $f(3/4) = 1$
 - III. $f(x) = xf(x)$
 - IV. $f(0) = f(1)$
- Then, the correct code is
- (a) only I is vector space
 - (b) only I and IV are vector spaces
 - (c) only I, III and IV are vector spaces
 - (d) All are vector spaces
44. Let $Q[\sqrt{3}] = \{a + b\sqrt{3} : a, b \in Q\}$, then which of the following is correct?
- (a) $Q[\sqrt{3}]$ is infinite dimensional vector space over R
 - (b) $Q[\sqrt{3}]$ is two-dimensional vector space over R
 - (c) $Q[\sqrt{3}]$ is two-dimensional vector space over Q
 - (d) R is finite dimensional vector space over $Q[\sqrt{3}]$

Answers Level-II

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (d) | 4. (d) | 5. (c) | 6. (b) | 7. (a) | 8. (a) | 9. (b) | 10. (a) |
| 11. (a) | 12. (b) | 13. (c) | 14. (b) | 15. (c) | 16. (d) | 17. (c) | 18. (b) | 19. (c) | 20. (c) |
| 21. (c) | 22. (a) | 23. (b) | 24. (b) | 25. (c) | 26. (b) | 27. (c) | 28. (d) | 29. (b) | 30. (c) |
| 31. (d) | 32. (d) | 33. (a) | 34. (d) | 35. (d) | 36. (d) | 37. (a) | 38. (b) | 39. (a) | 40. (d) |
| 41. (a) | 42. (b) | 43. (c) | 44. (c) | | | | | | |